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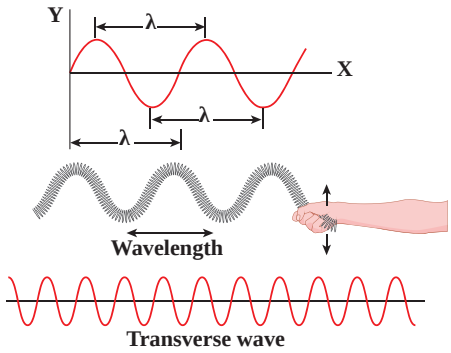


Figure 11.11 Wavelength for transverse waves

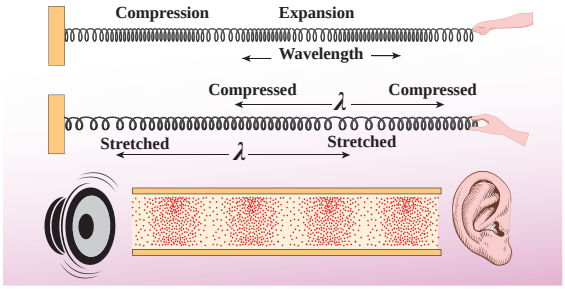
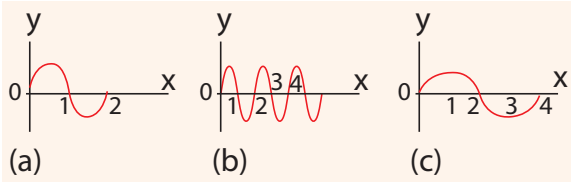


Figure 11.12 Wavelength for longitudinal waves

EXAMPLE 11.1

Which of the following has longer wavelength?



Answer is (c)

In order to understand frequency and time period, let us consider waves (made of three wavelengths) as shown in Figure 11.13 (a). At time $t = 0$ *s*, the wave reaches the point A from left. After time $t = 1$ *s* (shown in figure 11.13(b)), the number of waves which have crossed the point A is two. Therefore, the frequency is defined as "the number of waves crossing a point per second" It is measured in *hertz* whose symbol is *Hz*. In this example,

$$f = 2 \text{ Hz} \qquad (11.1)$$

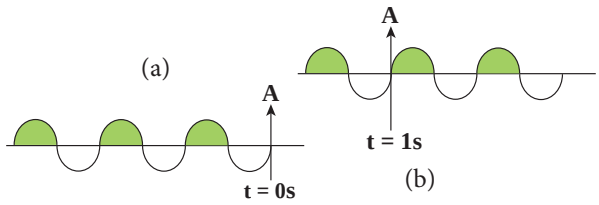


Figure 11.13 A wave consisting of three wavelengths passing a point A at time (a) $t = 0$ s and (b) after time $t = 1$ s

If two waves take one second (time) to cross the point A then the time taken by one wave to cross the point A is half a second. This defines the time period T as

$$T = \frac{1}{2} = 0.5 \text{ s} \qquad (11.2)$$

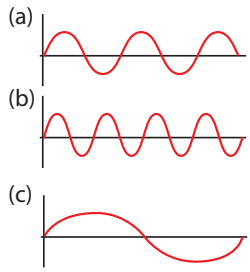
From equation (11.1) and equation (11.2), *frequency and time period are inversely related* i.e.,

$$T = \frac{1}{f} \qquad (11.3)$$

Time period is defined as *the time taken by one wave to cross a point.*

EXAMPLE 11.2

Three waves are shown in the figure below



Write down

(a) the frequency in ascending order

(b) the wavelength in ascending order

Solution

(a) $f_c < f_a < f_b$

(b) $\lambda_b < \lambda_a < \lambda_c$